

DAVID THANPHILOM

SENIOR FULL-STACK SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 8+ years of experience designing, architecting, and optimizing high-traffic web applications, automated data pipelines, and scalable digital systems. Expert at translating complex data contracts and workflows into high-performance, accessible user interfaces using React, TypeScript, and Node.js frameworks (Express). Proven track record of architecting serverless systems, integrating AI-driven API layers, designing relational data schemas, and deploying backend workflows directly to cloud infrastructure to handle high-density datasets.

TECHNICAL SKILLS

- **Languages & Frameworks:** Modern JavaScript, TypeScript, React, Next.js, Node.js, Express.js, PHP, HTML5, CSS3 (Tailwind CSS, Sass/Less)
- **API & Data Architecture:** GraphQL, REST APIs, Gemini API / Vertex AI, Custom Content Modeling, Relational Databases, Amazon S3, Shopify Storefront API, Liquid, WordPress
- **Tools & Workflows:** Vercel, AWS, Git/GitHub, Figma-to-Code, Rapid Prototyping, AI-Assisted Development, Core Web Vitals, Google Tag Manager (GTM), Agile/Scrum

PORTFOLIO PROJECTS

HEADLESS E-COMMERCE SHOPIFY ENGINE (NEXT.JS & GRAPHQL)

- Engineered a fully decoupled headless commerce application using Next.js client component hydration, interfacing natively with the Shopify Storefront API via highly optimized asynchronous GraphQL queries.
- Built an on-the-fly array processing engine using JavaScript matrix filtering (.filter) and real-time numerical sorting (.sort) configurations to handle complex price variations and out-of-stock exclusions smoothly without thread latency.
- Implemented a type-safe local storage state architecture (React Context API) with an optimistic UI cart drawer to handle zero-latency line migrations, instant subtotal aggregation, and automated radar-dot Toast notification alerts.
- Offloaded PCI payment compliance and transactional liabilities entirely by generating encrypted session checkout tokens via remote GraphQL mutations, establishing a seamless client redirect mesh.

REAL-TIME EVENT ORCHESTRATION ENGINE & KARAOKE HUB

- **WebSocket Synchronization:** Implemented Supabase Real-Time Client Pipelines utilizing PostgreSQL replication via WebSockets to stream instant queue lineup synchronization to client viewports simultaneously upon backend mutations.
- **Serverless Proxying & Debouncing:** Architected a secure Next.js Serverless API Route to act as a YouTube Data API v3 Proxy to preserve private credentials, integrating client-side 500ms debouncing optimization to drastically reduce API quota consumption during burst traffic.
- **Algorithmic Queue Sorting:** Programmed a high-performance queue-sorting layout built on the native HTML5 Drag and Drop API, leveraging mid-point float math database calculations ($\text{prev_order} + (\text{next_order} - \text{prev_order}) / 2$) to shift list elements instantly without forcing expensive row re-indexing.

- **Cross-Platform Product Engineering:** Implemented a unified sharing module utilizing the native Web Share API (navigator.share) for mobile device optimization, designing an asynchronous fallback to an external serverless QR generation microservice for desktop layout viewports.

WORK EXPERIENCE

SENIOR FULL-STACK DEVELOPER | TCWGLOBAL (CONTRACT) | JANUARY 2020 TO PRESENT

- **CI/CD & Production Deployment Ownership:** Fully owned the production deployment lifecycle and automated CI/CD release workflows via GitHub, managing secure serverless architectures and data-migration streams to cloud infrastructure.
- **Platform Engine Optimization:** Circumvented severe API platform memory caps (50-item retrieval ceiling) by engineering custom backend sub-pagination loops, unlocking seamless data rendering for 100+ documents on a single-page view.
- **Interactive Presentation Layers:** Engineered a custom front-end JavaScript engine using regular expressions to dynamically manipulate and format high-density multi-column datasets into highly readable, chronologically structured interface layers.
- **Database Architecture:** Designed a scalable relational database architecture utilizing nested content modeling constraints, successfully mapping complex object dependencies and reducing platform overhead.
- **Performance Engineering:** Leveraged Lighthouse and Chrome DevTools to optimize core asset delivery and client-side scripts, dramatically improving Core Web Vitals, driving mobile Largest Contentful Paint (LCP) down, and boosting storefront conversion rates (CRO) by 40%.
- **Frontend UI Leadership & Architecture:** Led team-wide frontend standards and architectural initiatives, establishing engineering guidelines for comprehensive **React.js** code reviews and scalable digital system delivery.

SENIOR WEB DEVELOPER | TEA COLLECTION | FEBRUARY 2017 TO JANUARY 2020

- **API & Serverless Architecture:** Engineered custom Node.js and Express backend architectures deployed on Vercel to establish stable, low-latency REST API endpoints.
- **Data Integration & AI:** Architected backend workflows to securely consume product catalogs and customer purchase data, integrating the Google Gemini API to deliver contextual product recommendations that drove personalization.
- **Cross-Functional Collaboration:** Collaborated remotely across distributed teams of project managers, designers, and marketers via Slack and Figma to transform high-fidelity wireframes into highly responsive web components.
- **Data-Driven Optimization:** Formulated complex A/B testing variations and implemented precise Google Tag Manager (GTM) event tracking to isolate user friction points, using live performance data to drive multi-brand conversion rate growth by 45%.
- **React.js & Architecture Ownership:** Owned complex modular frontend architectures using **React.js** and **TypeScript**, driving scalable development patterns, asynchronous peer reviews, and structured branch isolation loops.

FREELANCE FRONT-END ENGINEER | PONY RESCUE CLUB WEBSITE | MARCH 2026 – AUGUST 2026

- **Rapid Prototyping & Layouts:** Architected a fully responsive 7-page visual interface utilizing advanced CSS (Grid/Flexbox) layout paradigms, transforming data-heavy multi-column views into responsive mobile viewports.
- **Information Architecture:** Collaborated directly with business owners to transform rigid operational workflows into an intuitive interface using interactive web components and visual data cards.

FREELANCE WEB DEVELOPER | UPGROW INC | JULY 2021 TO AUGUST 2021

- **UI/UX Implementation:** Built highly semantic landing pages utilizing advanced layout frameworks, elevating touch target precision and accessibility for mobile interfaces.
- **Modular Systems:** Created reusable, cross-platform interactive marketing properties via semantic development methods, raising client engagement metrics by 35%.

WEB DEVELOPER (APPLEONE CONTRACT) | SAMTRANS & CALTRAIN | JUNE 2016 TO DECEMBER 2016

- **Interactive Mapping & SEO:** Configured complex transit datasets (GTFS real-time feeds) onto front-end maps while managing semantic SEO structural optimization across primary public transit interface.
- **Semantic Markup & Engineering:** Hand-coded responsive HTML, CSS, and vanilla JavaScript pages built directly from visual design specs, maintaining technical SEO structural compliance.

WEB DEVELOPER (AQUENT CONTRACT) | COST PLUS WORLD MARKET | MARCH 2015 TO NOVEMBER 2015

- **Enterprise Architecture:** Customized enterprise WordPress frameworks and built stable PHP server-side architectures engineered to support deep backend software upgrades.
- **Layout Customization:** Controlled advanced enterprise CMS utilities (MarketLive platform) to program, layout, and launch highly interactive web components and key promotional assets.

EDUCATION

WEB DEVELOPMENT CERTIFICATION | SEATTLE CENTRAL COLLEGE | 2014